# SIMATS ENGINEERING
## ASSESSMENT TOOL 1

**COURSE NAME: CSA0706 COURSE CODE: Computer Networks**

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**COURSE OUTCOME COVERED**

***CO1:*** *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

*Assessment Tool Weightage: 40%*

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Annexure A

**SHORT ANSWER TEST**

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***Code of Conduct:***

*I ____Maniyam Anushka(192525411)________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***

## OSI Model Numerical Problems - Detailed Assignment Solutions

### Question 1

Given: File=150

MB=150,000,000 bytes;

Bandwidth=50 Mbps;

Segment=1500 bytes;

DLL overhead=40 bytes/frame.

Formula: Segments = File Size / Segment Size.

Calculation: 150,000,000 / 1500 = 100,000 segments.

Formula: Frames = Segments = 100,000.

Formula: Overhead = Frames × 40 = 100,000 × 40 = 4,000,000 bytes.

Total transmitted = 150,000,000 + 4,000,000 = 154,000,000 bytes.

Bits transmitted = 154,000,000 × 8 = 1,232,000,000 bits.

Transmission time = 1,232,000,000 / 50,000,000 = 24.64 s.

Final Answer: 100,000 segments, 100,000 frames, 4,000,000-byte overhead, 24.64 s.

Explanation: Transport layer performs segmentation, sequencing, acknowledgements, retransmission and flow control. Data Link layer performs framing, MAC addressing, CRC error detection and reliable node-to-node delivery.

### Question 2

Given: Window size=6,

Total frames=48,

Frame size=1024 bytes.

Formula: Windows = Total Frames / Window Size = 48/6 = 8.

Retransmissions = 1 lost frame × 8 windows = 8.

Final Answer: Windows = 8; Retransmissions = 8.

Explanation: Flow control matches sender speed with receiver capacity, preventing buffer overflow, reducing congestion and improving throughput.

## Question 3

Given: Packet=12,000 bytes,

MTU=1500 bytes,

IP header=20 bytes.

Payload per fragment = 1500 − 20 = 1480 bytes.

Fragments = ceil(12000/1480)=9.

Header overhead = 9 × 20 = 180 bytes.

Final Answer: 9 fragments; 180-byte overhead.

Explanation: The Network layer fragments oversized packets, adds identification and fragment offset fields, routes fragments, and the destination reassembles them.

## Question 4

Given: Window size=6,

Total frames=48,

Frame size=1024 bytes.

Formula: Windows = Total Frames / Window Size = 48/6 = 8.

Retransmissions = 1 lost frame × 8 windows = 8.

Final Answer: Windows = 8; Retransmissions = 8.

Explanation: Flow control matches sender speed with receiver capacity, preventing buffer overflow, reducing congestion and improving throughput.

## Question 5

Given: File=600 MB; checkpoint every 60 MB; failure at 390 MB.

Checkpoints before failure: 60,120,180,240,300,360 = 6 checkpoints.

Data retransmitted = 390−360 = 30 MB.

Final Answer: 6 checkpoints; 30 MB retransmitted.

Explanation: Session-layer synchronization checkpoints allow communication to resume from the latest checkpoint instead of restarting.

### Question 6

Compressed size = 900 × (100−40)% = 900 × 0.60 = 540 MB.

Encryption overhead = 540 × 5% = 27 MB.

Encrypted size = 540 + 27 = 567 MB.

Reduction = (900−567)/900 ×100 = 37%.

Final Answer: Compressed=540 MB; Encrypted=567 MB; Reduction=37%.

Explanation: Presentation layer compresses data to reduce bandwidth and encrypts it for confidentiality.

### Question 7

Given: 3 GB ≈ 3000 MB;

Request size=3 MB;

Throughput=120 Mbps.

Requests = 3000/3 = 1000.

Data = 3000×8 = 24,000 Mb.

Time = 24,000/120 = 200 s.

Final Answer: 1000 requests; 200 seconds.

Explanation: Application layer provides FTP services, authentication, file transfer control and user interface.

### Question 8

Processing delay = 4+3+5+2+1 = 15 ms.

End-to-end delay = Processing + Propagation + Transmission = 15+18+12 = 45 ms.

Final Answer: 45 ms.

Explanation: Each OSI layer performs processing before forwarding data; propagation is travel time and transmission is bit sending time.

### Question 9

Useful payload/frame = 1600−80 = 1520 bytes.

Frames = ceil(80,000,000/1520)=52,632.

Overhead = 52,632×80 = 4,210,560 bytes.

Efficiency = 80,000,000/(80,000,000+4,210,560)×100 ≈95.0%.

Final Answer: 52,632 frames; 4,210,560-byte overhead; 95.0% efficiency.

Explanation: Protocol overhead reduces effective bandwidth but enables addressing, routing, sequencing and error control.

## Question 10

Compressed size = 240×0.70

= 168 MB

=168,000,000 bytes.

Segments =168,000,000/1400 =120,000.

Overhead =120,000×60 =7,200,000 bytes =7.2 MB.

Final transmitted =168+7.2 =175.2 MB.

Final Answer: Compressed=168 MB; Segments=120,000; Overhead=7.2 MB; Total=175.2 MB.

Explanation: Presentation compresses, Transport segments and ensures reliability, Network routes, Data Link frames and detects errors, Physical transmits bits.